8/9/25 TASK 5:-

JOIN QUIRES, EQUIVALENT AND RECURSIVE QUIRES

Aim:- To implement and execute joins, equivalent Queries and Recursive Queries in SQL.

Procedure:-

1. Create table DEPARTMENT & STUDENT 4.
2. Insert the values into tables
3. Perform join operation.
4. Perform Equivalent & Recursive Query.
5. Display result.

```
CREATE TABLE DEPARTMENT4(
        DEPTID INT PRIMARY KEY;
        DEPTNAME VARCHAR(50));
CREATE TABLE STUDENT4(
        STU_ID INT PRIMARY KEY,
        NAME VARCHAR(50),
        AGE INT,
        DEPTID INT,
        FOREIGN KEY (DEPT ID)
        REFERENCES DEPARTMENT4(DEPT ID)
        );
INSERT INTO DEPARTMENT4 VALUES
    (201, 'Computer Science'),
    (202, 'Electronics'),
    (203, 'Mechanical'),
INSERT INTO STUDENT4 VALUES
    (1, 'Ravi', 20, 201),
    (2, 'Sneha', 22, 201),
    (3, 'Amit', 19, 202),
```

(4, 'Priya', 24, 203),
(5, 'kiran', 23, 201);

SELECT FROM DEPARTMENT 4;

S.NO	DEPTID	DEPT NAME
1	201	Computer Science
2	202	Electronics
3	203	Mechanical.

SELECT* FROM STUDENT 4;

S.NO	STU-ID	NAME	AGE	DEPT ID
1	1	RAVi	20	201
2	2	Sneha	22	201
3	3	Amit	19	202
4	4	Priya	24	203
5	5	kiran	23	201

```
SELECT S.NAME, S.AGE, D.DEPTNAME
FROM STUDENT4 S
INNER JOIN DEPARTMENT4 D
ON S.DEPT ID = D.DEPT ID;
-- INNER JOIN
```

S.NO	NAME	AGE	DEPT NAME
1	RAVI	20	Computer Science
2	Sneha	22	Computer Science
3	Amit	19	Electronics
4	Priya	24	Mechanical
5	kiran	23	Computer Science.

-- LEFT OUTER JOIN

SELECT S. NAME, S. AGE, D. DEPT NAME
FROM STUDENT 4 S
LEFT JOIN DEPARTMENT 4 D
ON S. DEPT ID = D. DEPT ID;

S.NO	NAME	AGE	DEPT NAME
1	Ravi	20	Computer Science.
2	Sneha	22	Computer Science.
3	Amit	19	Electronics
4	Priya	24	Mechanics
5	Kiran	23	Computer Science

SELECT S. NAME, S. AGE, D. DEPARTNAME
FROM. STUDENT 4 S.
RIGHT JOIN DEPARTMENT 4 D
ON S. DEPT ID = D. DEPT ID;

S.NO	NAME	AGE	DEPT NAME
1	Ravi	20	Computer Science.
2	Sneha	22	Computer Science.
3	Kiran	23	Computer Science.
4	Amit	19	Electronics.
5	Priya	24	Mechanical.

SELECT TOP 3 S. NAME, S. AGE, D. DEPT NAME
FROM STUDENT 4 S
Full OUTER JOIN DEPARTMENT 4 D
ON S. DEPTID = D. DEPTID;

S.NO	NAME	AGE	DEPT NAME
1	Ravi	20	Computer Science
2	Sneha.	22	Computer Science
3	Amit	19	Electronics.

```
---E Quivalent QUERIES
---USING join
SELECT S.NAME, S.AGE
FROM STUDENT 4 S
JOIN DEPARTMENT 4D ON S.DEPT ID = D.DEPT ID
WHERE D.DEPT NAME = 'Compuler Science;
```

S.NO	NAME	AGE
1.	Ravi	20
2.	Sneha	22
3.	kiran	23

```
--RECURSIVE QUERIES
WITH COUNT CTE AS (
    SELECT 1 AS N
    UNION ALL
    SELECT N+1
    FROM COUNTCTE
    WHERE N<5
    )
SELECT * FROM COUNT CTE;
```

S.NO	N
1	1
2	2
3	3
4	4
5	5

EX NO	5
PE...	5
RESULT...	5
VIVA...	5
RECORD...	
TOTAL	
SIGN W... DATE	

<u>Result</u>:- Thus, implementation of Join, Quires, equivalent And Recursive Quires has successfully excuted And verified.